

Security+ 4.7 · Scripting & Automation

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.7** —
Scripting & Automation (8:20) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Automation: patching, log collection, provisioning, alert response. Python/Bash/PowerShell. Caution: scripts are also attack surface — use credential VAULT, code review, scope, library SCA.

High-Yield Core Concepts Tested on Exam

- Automation reduces human error — the main source of breaches
- Caution: script with hardcoded credentials = risk
- Use credential vaults for secrets in scripts
- Code review and scope limitation are essential
- SCA (Software Composition Analysis) for third-party libraries

Exam Tips & Classic Scenarios

- 'eliminates human error from repetitive tasks' = automation
- 'hardcoded credentials in script' = vulnerability
- 'automated patching' = reduces attack surface

Typical Exam Scenario: 'script runs patching on 200 servers using vault credentials' = secure practice.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.7 — Scripting & Automation)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. An attacker tests leaked passwords from one site on MULTIPLE other sites (password reuse):

- A)** Credential stuffing
- B)** Pure brute force
- C)** Rainbow table
- D)** Vishing

2. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'Automation reduces human error — the main source of breach'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: Automation reduces human error — the main source of breach

3. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'Caution: script with credential in code = risk'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Caution: script with credential in code = risk...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'eliminates human error from repetitive tasks'=automation. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'eliminates human error from repetitive tasks'=automation
- D)** Allow shared access among administrators with generic credentials

5. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'script runs patching on 200 servers with credentials from the vault' = secure practice. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'script runs patching on 200 servers with credentials from the vault' = secure practice
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

6. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'Automation reduces human error — the main source of breaches'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Automation reduces human error — the main source of breaches...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'Caution: script with credentials in code = risk'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Caution: script with credentials in code = risk...
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'eliminates human error from repetitive tasks'=automation'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'eliminates human error from repetitive tasks'=automation...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

9. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'script runs patching on 200 servers with vault credential' = secure practice. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'script runs patching on 200 servers with vault credential' = secure practice

10. Corporate scenario (Scripting & Automation): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.7), an analyst evaluates the following technical requirement: 'Automation reduces human error — the main source of breach.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Automation reduces human error — the main source of breach...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — Credential stuffing.

✓ **Why it's correct:** Credential stuffing is a cyberattack in which the attacker uses lists of credentials (username and password pairs) obtained from data breaches on one site to attempt authentication on multiple other sites, exploiting the common practice of password reuse. The attack is automated and relies on large volumes of leaked credentials. It differs from pure brute force because it does not attempt to guess passwords but uses already known combinations. It is a topic in the Threats, Vulnerabilities, and Mitigations domain of Security+ (SY0-701), objective 2.4, which covers attack techniques.

✗ **Why the others are wrong:**

- *Pure brute force:* Pure brute force is an attack method that exhaustively tries all possible character combinations until it finds the correct password. In credential stuffing, the attacker does not test random combinations; they use credentials previously leaked and valid in another context. Brute force is computationally intensive and easily detected by account lockouts, while credential stuffing exploits password reuse and can be stealthier because the credentials are legitimate. Therefore, pure brute force does not describe the scenario of testing leaked passwords on multiple sites.
- *Rainbow table:* A rainbow table is a hash-cracking technique that uses precomputed tables of password hashes to reverse cryptographic hash functions. It is used when the attacker obtains password hashes and wants to discover the plaintext passwords. However, the statement describes the use of already leaked passwords (in plaintext or already cracked) to authenticate on other sites, not hash cracking. Rainbow tables do not directly apply to credential stuffing, which does not require breaking encryption but rather exploiting credential reuse.
- *Vishing:* Vishing is a form of social engineering that uses phone calls or voice messages to deceive victims and obtain confidential information, such as passwords or financial data. It does not involve automated testing of leaked credentials across multiple sites. The described scenario is clearly a technical automation attack, not a human persuasion interaction. Therefore, vishing does not apply to the context of credential stuffing, which is a mass attack based on credential lists.

2) D — Implement the corresponding solution: Automation reduces human error — the main source of breach.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): Automation reduces human error — the main source of breach.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

3) B — Implement the corresponding solution: Caution: script with credential in code = risk....

✅ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): Caution: script with credential in code = risk. Hardcoding credentials in scripts exposes them to anyone with access to the code, leading to credential leakage and unauthorized access. Best practices include using secure secret management, environment variables, or vaults.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) C — Implement the corresponding solution: 'eliminates human error from repetitive tasks'=automation.

✅ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): 'eliminates human error from repetitive tasks'=automation.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) A — Implement the corresponding solution: 'script runs patching on 200 servers with credentials from the vault' = secure practice.

✅ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): 'script runs patching on 200 servers with credentials from the vault' = secure practice. Using a vault for credentials ensures secure storage and access, which is a best practice for automation.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: Automation reduces human error — the main source of breaches....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): Automation reduces human error — the main source of breaches. Automation ensures consistent execution of security tasks, eliminating mistakes caused by fatigue, oversight, or lack of expertise.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) C — Implement the corresponding solution: Caution: script with credentials in code = risk....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): Caution: script with credentials in code = risk. Hardcoding credentials in scripts is a dangerous practice because it exposes sensitive information to anyone who can read the script, including attackers who gain access to the system or repository. Best practices include using secure secret management solutions (e.g., vaults, environment variables, or managed identities) to store and retrieve credentials securely.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) A — Implement the corresponding solution: 'eliminates human error from repetitive tasks'=automation....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): 'eliminates human error from repetitive tasks'=automation. Automation uses scripts and tools to perform repetitive tasks consistently, reducing the risk of human error, improving efficiency, and ensuring compliance with security policies.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks,

injections, or runtime malware.

- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: 'script runs patching on 200 servers with vault credential' = secure practice.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): 'script runs patching on 200 servers with vault credential' = secure practice. Using a vault (e.g., HashiCorp Vault, Azure Key Vault) to manage credentials for automated patching ensures secrets are not hardcoded, rotated, and audited, reducing risk of credential theft.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) B — Implement the corresponding solution: Automation reduces human error — the main source of breach....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.7 (Scripting & Automation): Automation reduces human error — the main source of breach. Automation enforces consistent, repeatable processes, eliminating manual mistakes such as misconfigurations, forgotten patches, or accidental data exposure, which are leading causes of security incidents.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.